

ECON 311

Lecture 6: Solow Growth Model

The Solow Growth Model

Introduction

- The production model is very useful because is able to display two features of the data:

1. $\frac{y_{ind}}{y_{nind}} = \frac{\bar{A}_{ind}(\bar{k}_{ind})^{\frac{1}{3}}}{\bar{A}_{nind}(\bar{k}_{nind})^{\frac{1}{3}}}$ (differences in levels of y)

2. $g_{y_{ind}} - g_{y_{nind}} = (g_{\bar{A}_{ind}} + \frac{1}{3}g_{\bar{k}_{ind}}) - (g_{\bar{A}_{nind}} + \frac{1}{3}g_{\bar{k}_{nind}})$ (differences in growth rates of y)

Growth Algebra

- Nevertheless, it does not provide any explanation as to why the growth rates are so different across countries.
 - ▶ The Solow Model provides an explanation for K accumulation, which will help us to explain why capital per worker grows at a faster pace in some countries.
 - ▶ As we have seen this is only a part of the explanation, but it is a very useful starting point for understanding the big differences in growth rates we observe in the data.

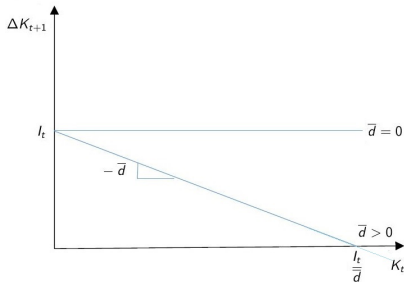
Capital Accumulation

- At the center of the Solow Model is the capital accumulation equation:

$$K_{t+1} = I_t + (1 - \bar{d})K_t$$

- ▶ I =investment, $\bar{d}$ =depreciation rate ($0 \leq \bar{d} \leq 1$)
- We can rewrite this equation in changes: $\Delta K_{t+1} = K_{t+1} - K_t$:

$$\Delta K_{t+1} = I_t - \bar{d}K_t$$



Capital Accumulation

- Investing in capital for $t + 1$ requires resources at t :

$$C_t + I_t = Y_t \quad (\text{resource constraint})$$

- ▶ An assumption of the Solow model is that capital and consumption goods are produced using the same production function.
- We are going to assume that individual decisions to save and invest is exogenous:

$$I_t = \bar{s}Y_t \quad (\text{allocation of resources}) \quad , \quad 0 \leq \bar{s} \leq 1$$

- Combining the allocation of resources with the capital accumulation equation, we obtain:

$$\Delta K_{t+1} = \bar{s}Y_t - \bar{d}K_t \quad (\text{law of motion})$$

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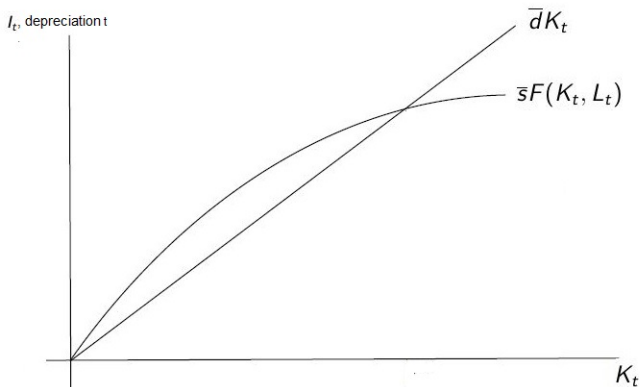
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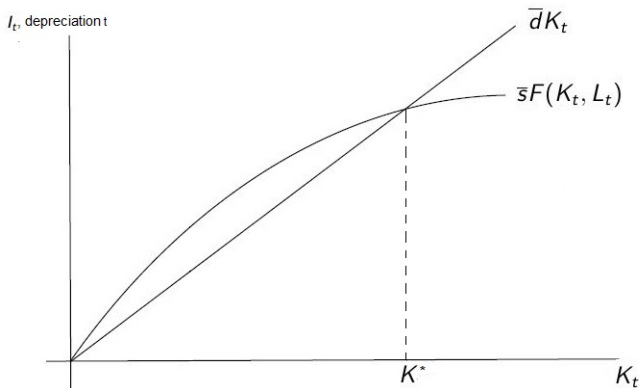
$$\Delta K_{t+1} = \bar{s}F(K_t, L_t) - \bar{d}K_t \quad (\text{law of motion})$$

Dynamics of Capital Accumulation



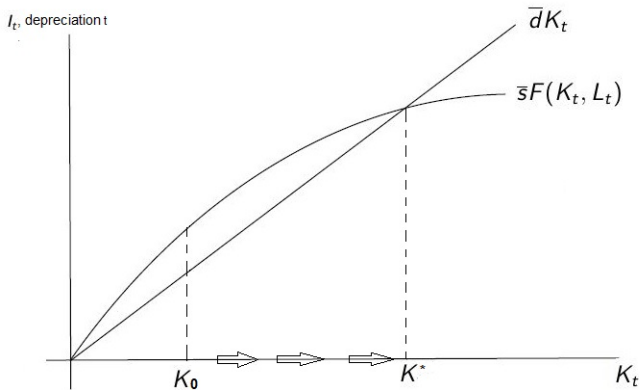
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Dynamics of Capital Accumulation



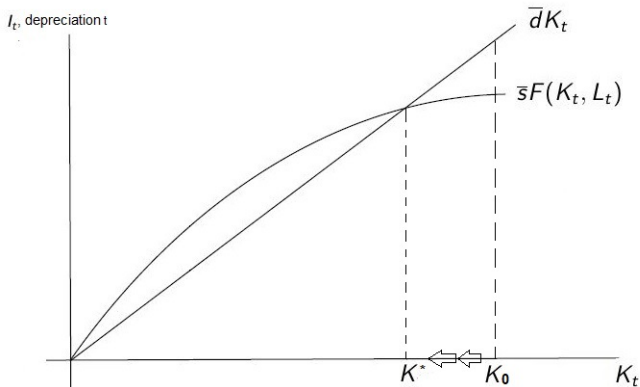
$$\underbrace{\Delta K_{t+1}}_{K_{t+1}=K_t} = 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} = \bar{d}K_t$$

Dynamics of Capital Accumulation



$$\underbrace{\Delta K_{t+1}}_{K_{t+1} > K_t} > 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} > \bar{d}K_t$$

Dynamics of Capital Accumulation



$$\underbrace{\Delta K_{t+1}}_{K_{t+1} < K_t} < 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} < \bar{d}K_t$$

Dynamics of Capital Accumulation

- To simplify the analysis, we are going to assume that: $L_t = \bar{L}$
- At K^* , we say that the economy is at its steady-state equilibrium.
- We can solve for K^* as a function of exogenous variables and parameters:

$$K^* = \bar{L} \left(\frac{\bar{s}\bar{A}}{\bar{d}} \right)^{\frac{3}{2}}$$

or in per capita terms: $k^* = \left(\frac{\bar{s}\bar{A}}{\bar{d}} \right)^{\frac{3}{2}}$

- We are also able to solve for the steady-state level of output:

$$Y^* = \bar{L} \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{\bar{d}} \right)^{\frac{1}{2}}$$

or in per capita terms: $y^* = \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{\bar{d}} \right)^{\frac{1}{2}}$

Cross-country differences in $\bar{s}$ and growth

- Suppose there are two countries, CAN and ARG, with the same $\bar{A}$, $\bar{L}$, $\bar{d}$ and K_0 but with different $\bar{s}$

$$\bar{s}_{CAN} > \bar{s}_{ARG}$$

- At $t = 0$ both countries will have the same level of output per worker:

$$y_0 = \bar{A} \left(\frac{K_0}{\bar{L}} \right)^{\frac{1}{3}}$$

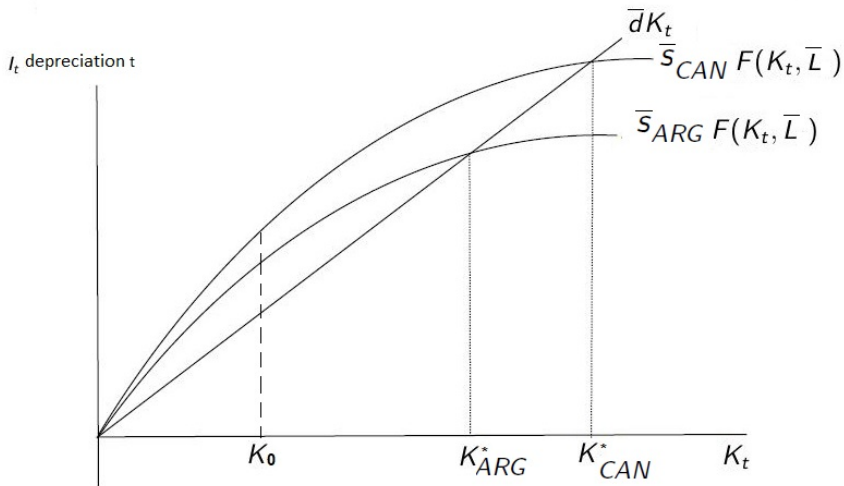
- However, in the long run:

$$y_{CAN}^* = \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}_{CAN}}{\bar{d}} \right)^{\frac{1}{2}} > \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}_{ARG}}{\bar{d}} \right)^{\frac{1}{2}} = y_{ARG}^*$$

Cross-country differences in $\bar{s}$ and growth

- Since the starting level of GDP per capita is the same but the level of GDP per capita is higher in Canada in steady state:
 - ▶ GDP per capita on average over the long run grows faster in Canada than in Argentina.
- The main reason for this difference in growth rates is that capital accumulation is higher in Canada.

Cross-country differences in $\bar{s}$ and growth



Solving for Equilibrium

- For any t , the aggregate economy in the Solow Model is characterized by five equations and five unknowns (K_t, L_t, I_t, C_t, Y_t):

$$\Delta K_{t+1} = \bar{s}F(K_t, L_t) - \bar{d}K_t \quad (\text{law of motion})$$

$$Y_t = \bar{A}K_t^{\frac{1}{3}}L_t^{\frac{2}{3}} \quad (\text{production function})$$

$$C_t + I_t = Y_t \quad (\text{resource constraint})$$

$$L_t = \bar{L} \quad (\text{labour force})$$

$$I_t = \bar{s}Y_t \quad (\text{allocation of resources})$$

- For an initial level of capital, K_0 , we can solve for the endogenous variables in any time period.
 - Instead of solving the model for any time period, which requires a lot of algebra, we will only solve analytically for the steady state (ss) and provide graphical interpretation for other time periods.

Appendix

Growth Algebra

- Algebraic properties of growth rates (let the variable i grow at a constant growth rate g_i):

$$z = \frac{x}{y} \quad \Rightarrow g_z = g_x - g_y$$

$$z = x \cdot y \quad \Rightarrow g_z = g_x + g_y$$

$$z = x^a \text{ (where } a \text{ is a constant)} \quad \Rightarrow g_z = a \cdot g_x$$

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